

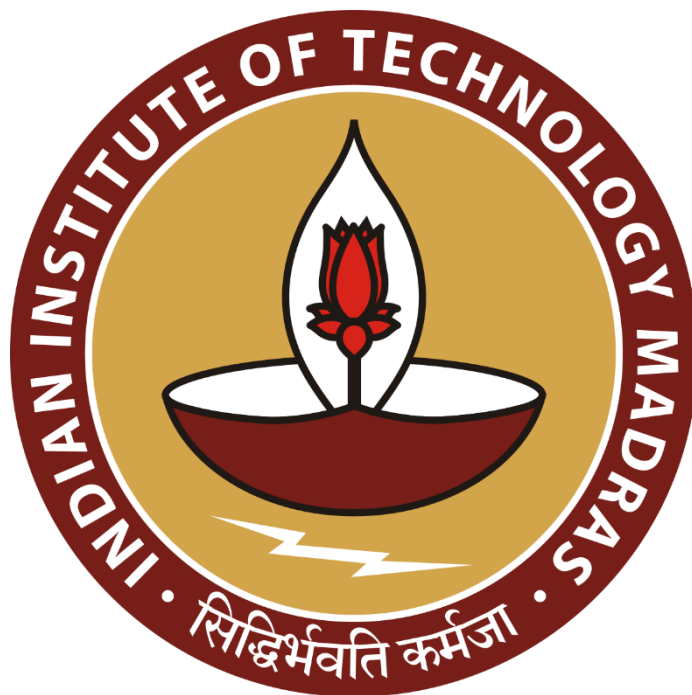
Inventory Management and Purchase Optimization Analysis for Rajesh Surgical

A Mid-Term report for the BDM capstone Project

Submitted by

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1 Executive Summary

The project titled “**Inventory Management and Purchase Optimization Analysis for Rajesh Surgical**” focuses on improving the company’s operational efficiency and profitability through a data-driven approach. Rajesh Surgical, a Moradabad-based wholesaler of surgical and medical supplies, acts as a key link between manufacturers and its network of hospitals, clinics, and retail outlets.

The company faces inefficiencies in its purchasing and inventory practices, leading to overstocking of low-demand items that tie up capital and understocking of high-demand products that cause missed sales opportunities. Additionally, the company’s current purchasing approach does not adequately address price fluctuations or cost-saving opportunities, creating scope for optimizing procurement timing and expenditure. To address these issues, purchase data from April 2024 to March 2025 was analyzed using Python and Microsoft Excel to classify inventory based on value and criticality and to provide a foundation for future forecasting and cost optimization.

The data was initially unstructured and required cleaning and transformation in Python to correct misaligned entries and standardize item names. Subsequent analysis in Excel included descriptive statistics and ABC–VED classification, revealing that nearly 20% of items contribute 80% of total purchase value. Further, VED analysis was applied to A-category items to classify them as Vital, Essential, or Desirable, based on their functional importance and role in business operations. This helped identify high-value and high-criticality items that require strict inventory control and continuous monitoring.

Further analysis on trend and seasonality patterns (using data from April 2022–March 2024) has been initiated to uncover periodic purchasing behaviors. Insights from this study will support the next phase—forecasting and purchase optimization—to establish an efficient, data-driven inventory strategy.

2 Proof of Originality of the Data

- **Business Name:** Rajesh Surgical
- **Address:** Kaji Ki Sarai, Kothiwalan, Moradabad (244001), Uttar Pradesh
- **Owner's Name:** Mr Ankit Panday
- Drive link to Letter from the organisation on letterhead with stamp and sign. : [Letter](#)
- Drive link to access original Raw Data: [Raw Data](#)
- Drive link to Images of Rajesh Surgical: [Images](#)
- Drive link to Video Interaction with Business Owner (Mr Ankit Panday): [Video Link](#)

3 Metadata and Descriptive Statistics

The dataset used for this project was obtained from the purchase records of Rajesh Surgical, covering a three-year period from April 2022 to March 2025. It contains day-wise purchase transactions, providing detailed information about each item purchased, its quantity, rate, and total cost.

The raw dataset originally included the following columns:

“Date”, “Bill”, “Item Description”, “Batch”, “Quantity”, “Free”, “Rate”, and “Amount”.

However, for the purpose of this analysis, the concerned columns are:

“Date”, “Item Description”, “Quantity”, “Rate”, and “Amount”, as these are directly relevant to understanding purchasing behavior, expenditure trends, and inventory classification.

Column Name	Description	Data Type	Relevance to Analysis
Date	Purchase transaction date	DateTime	Used for trend, monthly totals, and seasonality analysis
Item Description	Product name (standardized and cleaned)	Text/String	Used for ABC–VED classification
Quantity	Number of units purchased	Numeric	Used to calculate total purchase volume
Rate	Unit price of each product	Numeric (₹)	Used to derive purchase value
Amount	Total purchase value (Quantity × Rate)	Numeric (₹)	Used to compute item-wise and category-wise contribution

RAJESH SURGICAL							
KAJI KI SARAI, KOTHIWALAN, MORADABAD							
Phone : 9219537999, 7060280809 E-Mail : ANKITPANDAY139@GMAIL.COM							
Licence No. : UP2120B000921, UP2121B000917							
GSTIN : 09ASXPS6634D1Z1 TIN No. : 09359001820 Food Lic.No. : 22722722000007							
PURCHASE SUMMARY							
01-04-2024 - 31-03-2025							
Date	Bill	Item Description	Batch	Qty.	Free	Rate	Amount
01-04-2024	000001	DISPO. 3PLY	ELAS		35	85.7160	3000.06
01-04-2024	000001	DISPO. 3PLY	ELAS		240	75.0000	18000.00
01-04-2024	000001	SHOE COVER	PLAST		60	160.0000	9600.00
01-04-2024	000001	SAMA GUAZE	SMDGS3801		1800	11.1625	20092.50
01-04-2024	000001	ARC 4*3.6M		23072502	260	95.0000	24700.00
01-04-2024	000001	ARC 3*3.6M		23072502	201	85.0000	17085.00
01-04-2024	000001	CAP BUFFERANT			160	52.0000	8320.00
01-04-2024	000001	CISLON PLUS	100M AL2403050		200	9.0000	1800.00
01-04-2024	000001	CISLON PLUS	1000 AL2403046		45	44.0000	1980.00
01-04-2024	000001	HANDLOOM GAUGE	S 054		500	14.2880	7144.00
01-04-2024	119	AVIL INJ. 2ML		2123114	1800	4.4019	7923.48
	01-04-2024	2425/0002 BIOS DNS CP	500M CDS24029		1920	12.0920	23216.64
	01-04-2024	2425/0002 BIOS NS FFS	100M CNS24012		2000	6.6500	13300.00
	01-04-2024	2425/0002 BIOS NS CP	500ML CNB24070		1200	12.0920	14510.40
	01-04-2024	2425/0002 BIOS RL CP	500ML CRL24061		960	12.4600	11961.60
	01-04-2024	2425/0002 BIOS OFLO MONO	F OFF23040		2000	9.1000	18200.00
16		01-04-2024			13381		200833.68

(Raw Data Snapshot with all Columns)

Date	Item Description	QTY	Rate	Amount
01-04-2024	SHOE COVER PLAST	60	160	9600
01-04-2024	SAMA GUAZE	1800	11.1625	20092.5
01-04-2024	ARC 4*3.6M	260	95	24700
01-04-2024	ARC 3*3.6M	201	85	17085
01-04-2024	CAP BUFFERANT	160	52	8320
01-04-2024	CISLON PLUS 100M	200	9	1800
01-04-2024	CISLON PLUS 1000	45	44	1980
01-04-2024	HANDLOOM GAUGE S	500	14.288	7144
01-04-2024	AVIL INJ. 2ML	1800	4.4019	7923.48
01-04-2024	BIOS DNS CP 500M	1920	12.092	23216.64
01-04-2024	BIOS NS FFS 100M	2000	6.65	13300

(Cleaned Data Snapshot with relevant Columns)

The dataset contains day-wise purchase records over a three-year period, distributed as follows:

- April 2022 – March 2023 (Year 1) (9614 rows)
- April 2023 – March 2024 (Year 2) (11,119 rows)
- April 2024 – March 2025 (Year 3) (8227 rows)

For analytical purposes:

- The most recent year (April 2024–March 2025) is used for ABC and VED analysis, representing the company's current purchase and inventory behavior.
- The previous two years (April 2022–March 2024) will be used for trend and seasonality analysis, leveraging the day-wise data to identify cyclical purchase and price patterns.

Purchase Date-

This Column is used for trend, monthly totals, and seasonality analysis. The minimum purchase is made in December, and the Maximum purchase is made in May.

	Purchase
Max	₹ 1,63,51,522
Min	₹ 60,17,864
Average	₹ 94,53,752

Item Description -

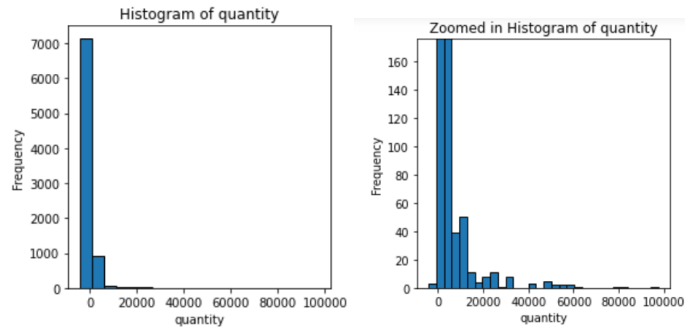
This Column gives the names of items purchased and is used for ABC–VED classification. There are a total of 1671 unique items. Out of which 327 contribute to 80% of the total purchase, 448 to 15% and 896 to 5% of the total purchase.

Quantity-

This column gives the number of units purchased for a particular item.

Most sales involve small quantities, with a few very large orders creating a right-skewed pattern — showing that bulk purchases are rare but significant when they occur.

Desc Stats	quantity
count	8,227.00
mean	751.60
std	3,394.14
min	-4,000.00
25%	12.00
50%	60.00
75%	400.00
max	97,500.00

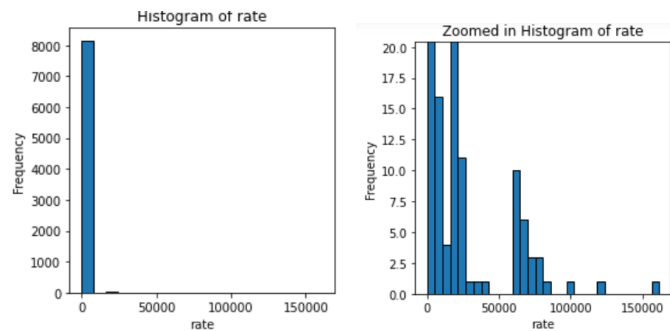


(Here, the -ve value signifies that some of the purchased items were cancelled or returned)

Rate-

This column gives the Unit price of each product.

Desc Stats	rate
count	8,227.00
mean	598.07
std	4,658.27
min	0.01
25%	18.19
50%	92.50
75%	269.82
max	1,61,600.00

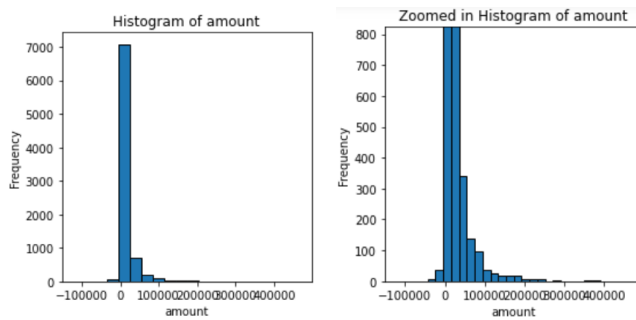


The rate distribution is multi-modal and right-skewed, indicating different pricing tiers — likely reflecting multiple product categories or types with varying price ranges.

Amount-

This column gives the Total purchase value (Quantity × Rate).

Desc Stats	amount
count	8,227.00
mean	13,789.35
std	26,311.66
min	-1,22,000.00
25%	2,232.38
50%	5,840.00
75%	14,732.16
max	4,68,072.00



(Here, the -ve value signifies that some of the purchased items were cancelled or returned)

Transaction amounts are highly right-skewed, meaning most transactions are of low value, but a small number of large transactions dominate total revenue.

4 Detailed Explanation of Analysis Process & Methods

Data Collection- The raw data was provided directly by Rajesh Surgical as three separate Excel files, covering April 2022–March 2023, April 2023–March 2024, and April 2024–March 2025. Each file

contained day-wise purchase transactions. The dataset included the columns Date, Bill, Item Description, Batch, Quantity, Free, Rate, and Amount. For this analysis, the relevant fields were identified as Date, Item Description, Quantity, Rate, and Amount, as they directly influence purchase value, demand, and pricing trends.

Data Preprocessing- The three files were merged into a single consolidated dataset. Several rounds of data cleaning and preprocessing were performed before analysis. Column names were standardized, and missing or misaligned values were corrected. Misalignment typically occurred when cells shifted left or right during data entry; these were fixed using pattern-based and manual checks. Item descriptions were standardized by converting them to uppercase and removing punctuation, ensuring that identical products with slightly different spellings were treated as one. Duplicate records were dropped, and invalid or summary rows such as “TOTAL” or “GRAND TOTAL” were removed.

Numeric columns (*Quantity*, *Rate*, *Amount*) were verified for consistency, ensuring all entries were numeric. Where possible, the “Amount” column was cross-validated as “Quantity x Rate” to detect anomalies. Items with significant discrepancies were flagged for manual review. The resulting cleaned dataset contained structured day-wise entries across all three financial years and was exported for further analysis in Excel.

Analysis Methods for identified problems-

The analysis was designed to address the following two key business problems faced by *Rajesh Surgical*:

1. Ineffective Inventory Planning & Suboptimal Purchase Strategy:
The company’s current purchasing and stocking practices result in overstocking of slow-moving items and understocking of high-demand products, leading to capital being locked in unsold inventory and potential revenue loss.
2. Price Optimization:
The company’s purchasing decisions do not consider price fluctuations or seasonal demand cycles, causing missed opportunities for bulk procurement at favorable rates.

1. Descriptive Statistics-

Descriptive analysis was conducted to understand the distribution and variability of purchasing data. Measures such as mean, median, standard deviation, and percentiles were calculated for *Quantity*, *Rate*, and *Amount*. This helped identify spending concentration, high-value products, and potential outliers. The analysis provided the foundation for later classification and confirmed that a small percentage of items account for a large share of total purchases—an early indication of a Pareto pattern.

2. ABC Analysis-

ABC analysis was applied to the most recent year (April 2024–March 2025) to prioritize items by their contribution to total purchase value. Items were ranked by purchase amount and categorized as:

- **A:** top 20% of items contributing around 80% of total value,
- **B:** next 30% contributing about 15%,

- C: remaining 50% contributing the final 5%.

This classification highlights where management should focus attention and working capital. ABC analysis is particularly appropriate here because it simplifies decision-making for a large product base (over 1,600 items) and directly links to the overstocking problem—ensuring high-value products are neither over-purchased nor understocked.

3. VED Classification-

The A-category items were further classified using VED analysis—a method that categorizes products based on their operational criticality:

- V (Vital): items essential for daily operations,
- E (Essential): necessary but not immediately critical,
- D (Desirable): low-criticality or optional items.

Tagging was done using product names and expert input from the organization.

The ABC–VED combination is valuable because it addresses both financial value (ABC) and functional importance (VED). For example, an AV (A-Vital) item like *Infusion Sets* demands continuous availability, while an AD (A-Desirable) item like *Abdominal Belt* can be restocked less frequently. This matrix directly informs stocking and procurement priorities.

4. Trend and Seasonality Analysis-

The first two years of data (April 2022–March 2024) have been analyzed to identify seasonal and cyclical purchasing patterns. Monthly aggregations will be used to observe demand fluctuations and purchasing trends. Detecting seasonal peaks and troughs will allow the company to plan purchases more effectively, reducing last-minute orders and ensuring stock availability during high-demand periods.

5. Price Variability and Cost Optimization-

The data also revealed notable variation in product rates across months. The next phase involves computing the coefficient of variation of item prices to flag products with volatile pricing. For such items, bulk purchasing can be planned during historically low-price months to reduce overall acquisition costs. This analysis directly supports the second business problem of optimizing purchasing expenditure.

Why are these methods most appropriate

The chosen analytical methods—Descriptive Statistics, ABC Analysis, VED Classification, and Trend/Seasonality Study—are well-suited for this problem because they balance interpretability and business relevance. Unlike complex machine-learning models, these approaches provide clear, actionable insights that management can immediately apply. Together, they form a comprehensive framework: ABC–VED tackles inventory prioritization and stock balance, while trend and price analysis guide timing decisions for cost-efficient procurement.

5 Results & Findings

Descriptive Insights-

Descriptive analysis of the purchase data revealed high variability in purchase quantity, rate, and frequency across different items, suggesting inconsistent procurement patterns.

To illustrate this, an item-level examination was performed on *RMS Infusion Set* — a high-value and frequently purchased medical supply. Between April 2024 and March 2025, this item was purchased 20 times, with quantities ranging from 500 to 49,500 units per order. The purchase rate fluctuated between ₹9.15 and ₹11.33, reflecting a price variation of nearly 24% for the same product.

Date	Item Description	Quantity	Rate	Amount
06-06-2024	RMS INFUSION SET	20000	9.1512	183024
01-07-2024	RMS INFUSION SET	25000	9.1512	228780
24-07-2024	RMS INFUSION SET	10000	9.1969	91968.7
06-11-2024	RMS INFUSION SET	10000	9.1969	91968.7
02-12-2024	RMS INFUSION SET	5000	9.1969	45984.35
13-12-2024	RMS INFUSION SET	15000	9.1969	137953.05
23-01-2025	RMS INFUSION SET	5000	9.1969	45984.35
06-02-2025	RMS INFUSION SET	10000	9.1969	91968.7
03-03-2025	RMS INFUSION SET	15000	9.1969	137953.05
08-04-2024	RMS INFUSION SET	5000	9.2496	46248
10-04-2024	RMS INFUSION SET	20000	9.348	186960
15-04-2024	RMS INFUSION SET	500	9.456	4728
15-04-2024	RMS INFUSION SET	49500	9.456	468072
04-05-2024	RMS INFUSION SET	8000	9.456	75648
04-05-2024	RMS INFUSION SET	4000	9.456	37824
04-05-2024	RMS INFUSION SET	1000	9.456	9456
04-05-2024	RMS INFUSION SET	2000	9.456	18912
07-05-2024	RMS INFUSION SET	15000	9.456	141840
07-05-2024	RMS INFUSION SET	10000	9.456	94560
14-06-2024	RMS INFUSION SET	25000	9.456	236400
03-02-2025	RMS INFUSION SET	1500	11.33	16995

Such inconsistency in both order size and pricing indicates the absence of a structured reorder policy or long-term vendor rate contract. The day-wise transaction pattern further confirms irregular purchase intervals, showing that procurement decisions are reactive rather than forecast-based.

Overall, these descriptive findings highlight the need for a data-driven purchase planning approach to ensure stable supply, price consistency, and optimal inventory levels — forming the basis for subsequent ABC–VED and trend analyses.

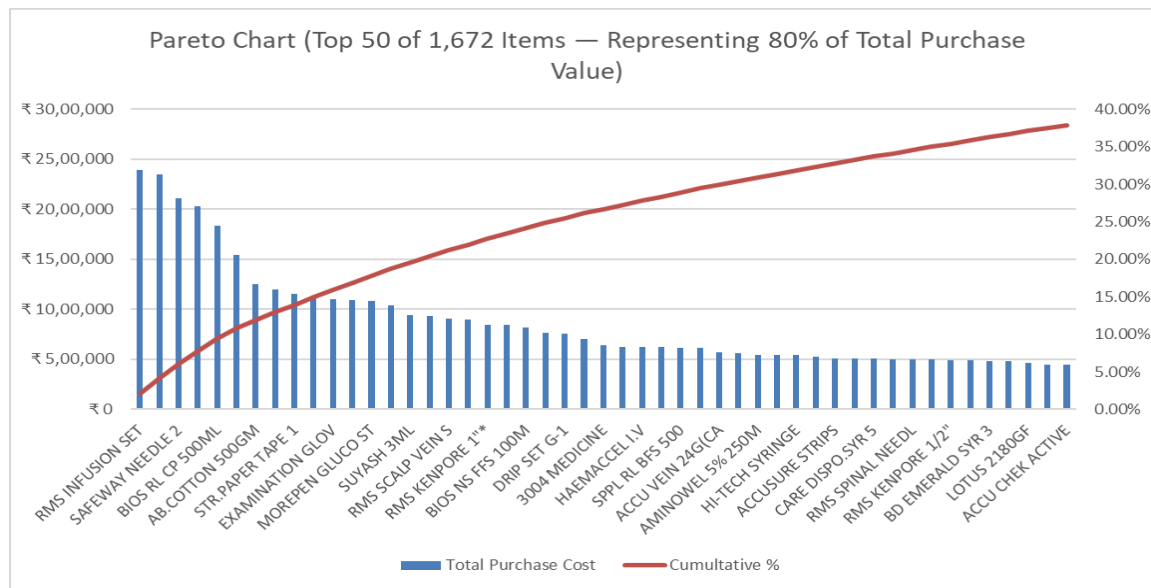
ABC Analysis

Using FY 2024–25 data, an ABC classification was conducted to categorize inventory based on total purchase value:

Category	No. of Items	% of Total Items	Total Purchase	Purchase %
A	327	20%	₹ 9,08,08,953	80%
B	448	27%	₹ 1,70,95,107	15%
C	896	54%	₹ 56,95,223	5%
Total	1671	100%	₹ 11,35,99,284	100%

This follows the Pareto principle, where 20% of items account for 80% of total purchases.

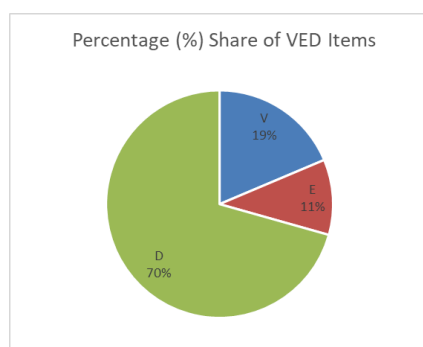
This result identifies the high-impact “A” items that should receive focused attention in procurement and inventory control to optimize cost and availability.



VED Classification-

To complement financial prioritization (ABC analysis) with operational importance, VED classification was applied to the 327 “A-category” items, identifying which are *Vital*, *Essential*, or *Desirable* for hospital operations.

Category	No. of Items	% of Total Items	Total purchase per Category	Purchase Share out of A	Purchase Share out of Total
V	61	19%	₹ 2,41,17,584	27%	21%
E	35	11%	₹ 1,02,48,520	11%	9%
D	231	71%	₹ 5,64,42,849	62%	50%
Total	327	100%	₹ 9,08,08,953	100%	80%



Category	V	E	D	Grand Total
A	61	35	231	327
B	44	33	371	448
C	27	34	835	896
Grand Total	132	102	1437	1671

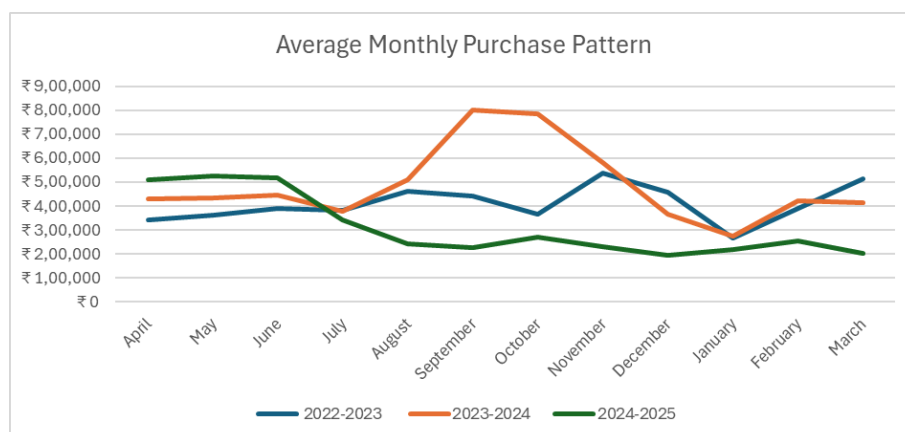
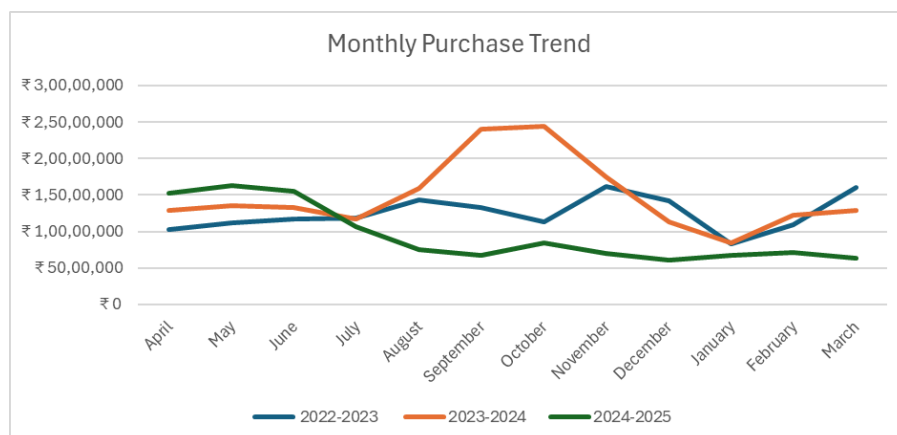
The Vital (V) items, though only 19% of A-category items, contribute 27% of A purchases and 21% of total spending, covering essential hospital supplies like infusion sets, syringes, and IV fluids that require uninterrupted availability.

The Essential (E) group forms 11% of A items and 9% of total spending, including frequently used items such as cotton and dressing pads.

The Desirable (D) items make up the majority (71%) and account for half of total purchase value, indicating a large number of low-priority consumables consuming significant expenditure.

Overall, the findings suggest that A–V items need tighter monitoring and vendor agreements to ensure availability.

Trend and Seasonality Analysis-



To explore purchasing behavior over time, monthly purchase totals were aggregated across three financial years (FY 2022–23, FY 2023–24, and FY 2024–25). Two charts are developed to visualize trends and seasonality.

Across FY 2022–23 to 2024–25, monthly purchase trends show clear seasonality.

- FY 2022–23 remained stable with mild peaks in November and March.
- FY 2023–24 exhibited a sharp rise during August–October, followed by a post-festive dip.
- FY 2024–25 reflected lower yet steady purchases.

The average monthly pattern across years confirms cyclical demand, with consistent spikes in Q2–Q3 (Aug–Oct) and March.

These insights indicate predictable seasonal behavior, supporting strategic bulk purchasing and rate optimization ahead of high-demand periods.